

Ganesh Gupta

817-822-1131 | ganeshg12ta@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Built **ML infrastructure** for production **model deployment** and **model evaluation**: a deployment-gated eval framework with automated regression testing that blocked **15%** of failed releases before production.
- Built **observability** and **autoscaling** for **vLLM** inference serving on multi-cloud **GPU** fleets (**AWS**, **GCP**, **CoreWeave**) via **Prometheus/Grafana**, correlating compute and network telemetry to catch performance regressions.
- Designed and scaled a pluggable **distributed executor** routing pipeline workloads across **Docker** and **AWS Lambda** with an **S3**-backed artifact store, eliminating redundant recomputation.
- Built multi-agent **GenAI** workflows on the **OpenAI Agents SDK** and **MCP** across **GPT**, **Claude**, **Llama**, and **Gemini**, tuning model choice per use case for quality and cost at **99%+ uptime**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Owned **ML infrastructure** end to end, from model deployment through evaluation, optimization, data processing, and debugging, cutting **p99 latency 420ms to 140ms** across **50K+ daily requests** on **GCP Cloud Run**.
- Optimized LLM serving with **kernel-level CUDA/Triton** work, **distributed KV cache**, **speculative decoding**, and **TensorRT-LLM** alongside **vLLM**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Designed **Python/C++** model-serving infrastructure with async request handling, batching, and automated failure recovery, reaching **99.9% availability** and cutting p99 latency **32%**.
- Profiled compute and memory bottlenecks** across Python/C++ inference services and optimized batching, tensor movement, and concurrency, raising throughput **2.4x** and cutting **GPU memory 27%**.

PROJECTS

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production **LLM evaluation** platform with rubric-based scoring, drift detection, and latency tracking across model versions; **FastAPI**, **PostgreSQL**, automated regression alerting.

LangFetch

tinyurl.com/LangFetch

- LangGraph** agent decomposing natural-language questions into dependency-ordered SQL subqueries with hybrid retrieval and re-ranking; deployable as an **MCP** server.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent** incident-diagnosis system orchestrating specialized tools to collect evidence, analyze production failures, and synthesize root-cause reports.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, C++, Java, Go, SQL

ML Infrastructure: Model Deployment, Model Evaluation, Optimization, Data Processing, Debugging, vLLM, TensorRT-LLM, CUDA/Triton

ML & AI: PyTorch, Multi-GPU Training, Reinforcement Learning, Quantization, RAG, LangGraph, MCP, OpenAI Agents SDK

Cloud: GCP (Vertex AI, Cloud Run, BigQuery), AWS (SageMaker, Lambda, S3), Kubernetes, Docker, Prometheus/Grafana

Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks

Certifications: AWS Certified Developer Associate, Google Cloud Professional Machine Learning Engineer